

# Cascade-Direction Inference on a T-cell Maturation Axis

A recovered timeline, a clean reversal, and a falsified hypothesis

cytokine-MIL project (§32) — session summary

2026-06-22

## Abstract

We extended the validated `cross_asym` cascade-*direction* statistic to a **T-cell maturation** axis (naive→effector→memory) on a public SARS-CoV-2 vaccination whole-PBMC CITE-seq atlas. The method **recovers the vaccination *timeline*** (D2→D11→D28) from one cross-sectional snapshot, but **exactly *reverses the differentiation state order***. A control-swap decomposition shows the reversal is *control-independent* and **falsifies our own first-pass explanation**. The insight: `cross_asym` reads a program-*acquisition* asymmetry while differentiation is a program-*retention* asymmetry — so it is correctly signed on a temporal/progression axis but mis-signed on a cell-state axis.

## 1 Setup

The direction statistic, for an ordered pair  $(a, b)$  with per-condition discovered signatures  $S_a, S_b$  and a control population, is

$$\text{cross\_asym}(a, b) = \underbrace{[s(a, S_b) - s(b, S_a)]}_{\text{raw cross-engagement (b)}} - \underbrace{[\text{ctrl}_{S_b} - \text{ctrl}_{S_a}]}_{\text{control term (a)}}, \quad (1)$$

where  $s(x, S)$  is the mean expression of gene set  $S$  in population  $x$ . The biology it encodes: an *upstream* stimulus’s cells carry *both* their own and the (autocrine) *downstream* program, so  $s(\text{up}, S_{\text{down}}) > s(\text{down}, S_{\text{up}})$ .

**Experiment.** Dataset: multimodal SARS-CoV-2 vaccination atlas (*Nat. Immunol.* 2023; Zenodo 7555405, no-auth). Whole-PBMC CITE-seq, days 0/2/11/28, 6 donors, 61,911 T cells. We reused the §30 disease-progression pipeline verbatim and ran two framings: **(i) STATE** — condition = maturation state {Naive, Effector, Memory} from CITE *surface protein* gating (CD45RO/CD27; RNA-independent, so state labels cannot leak into the RNA signatures), control = day-0 “Resting”; and **(ii) TIMEPOINT** — condition = {D2, D11, D28}, control = D0. An independent synthetic *apparatus* provides a distinct-program go/no-go gate.

## 2 Results: a recovered timeline and a reversed state order

The three framings give a sharp, internally-consistent picture (Table 1): the apparatus passes its hard gate, the *timeline* direction is recovered perfectly, and the *state* direction is exactly inverted. The tell-tale is the relationship between the direction statistic and its symmetric control (Fig. 1): on the apparatus and timepoint axes `cross_asym`  $\gg$  `directional_score`, but on the state axis the two *swap* (control 100%, direction 0%) — the “no-seed/monotone” regime the apparatus reproduces synthetically.

| framing                           | cross_asym acc. | symm. control | Kendall $\tau$ | verdict            |
|-----------------------------------|-----------------|---------------|----------------|--------------------|
| Apparatus (distinct-program gate) | 100%            | 50%           | +1.0           | PASS               |
| <b>Timepoint</b> D2<D11<D28       | <b>100%</b>     | 0%            | +1.0           | AMBER <sup>†</sup> |
| <b>State</b> Naive<Eff<Mem        | <b>0%</b>       | 100%          | -1.0           | RED (reversed)     |

Table 1: Direction recovery under the two real framings plus the synthetic gate. <sup>†</sup>Timepoint is AMBER *only* because the 6-donor bootstrap CI is wide ( $[0.33, 1.0]$ ); the point estimate is perfect. Recovered orders: timepoint = D2→D11→D28 (truth); state = Memory→Effector→Naive (the exact reverse of truth, hence  $\tau = -1.0$ ).

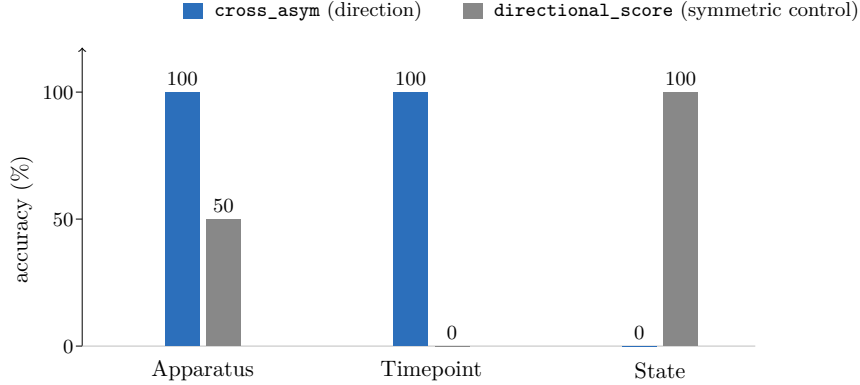


Figure 1: The “inverse fingerprint.” On the apparatus and timepoint axes the antisymmetric direction statistic beats its symmetric control (the healthy cytokine/COVID pattern). On the state axis the two swap places — direction 0%, control 100% — the hallmark of an axis where `cross_asym`’s core assumption runs backwards.

### 3 Anatomy of the reversal (and a falsified hypothesis)

Our first-pass explanation was that the control (day-0 “Resting”)  $\approx$  Naive, so the upstream state coincided with the baseline. To test it we ran a **control-swap decomposition**: holding the discovered signatures *fixed*, we varied only the control term in Eq. (1) — including a **ZERO** control (no subtraction = the raw cross-engagement, i.e. effect (b) alone). Table 2 and Fig. 2 are decisive: **the raw engagement is the *most* inverted** (+0.588 vs +0.140 under Resting), the control *reduces* the inversion, and *no* control choice un-inverts the Naive pairs. The baseline-coincidence hypothesis is **falsified**: effect (a) mitigates, it does not cause.

**The mechanism.** `cross_asym` assumes program *acquisition*: an upstream cell *gains* the downstream program (the cytokine-autocrine logic). Differentiation is the opposite — a program-*retention* asymmetry: the *mature* cell *retains* the *progenitor’s* program (memory T cells re-express the naive program: *IL7R*, *TCF7*, *SELL*, *CCR7*). Hence  $s(\text{Memory}, S_{\text{Naive}}) \gg s(\text{Naive}, S_{\text{Memory}})$  and the statistic faithfully points mature→naive. This is not a bug (the apparatus distinct-gate passes) and not fixable by control choice; it is the method meeting an axis whose biology is sign-flipped relative to its assumption.

| control                                | <code>cross_asym</code> (Eff,Nai) | <code>cross_asym</code> (Mem,Nai) | $\tau$ |
|----------------------------------------|-----------------------------------|-----------------------------------|--------|
| Resting (original)                     | +0.107                            | +0.140                            | -1.0   |
| <b>ZERO</b> = raw engagement (b)       | <b>+0.253</b>                     | <b>+0.588</b>                     | -1.0   |
| Balanced (removes naive/mem imbalance) | +0.121                            | +0.146                            | -1.0   |
| Naive as control                       | +0.138                            | +0.147                            | -1.0   |
| Memory as control                      | +0.159                            | +0.161                            | -1.0   |

Table 2: Control-swap decomposition (`analyze_vaccine_state_control_decomp.py`). The correct sign for an  $X$ -Naive pair is *negative* (Naive upstream); every control yields *positive* (inverted). The raw cross-engagement (ZERO) is the most inverted, so the inversion is intrinsic to the axis, not an artifact of the control.

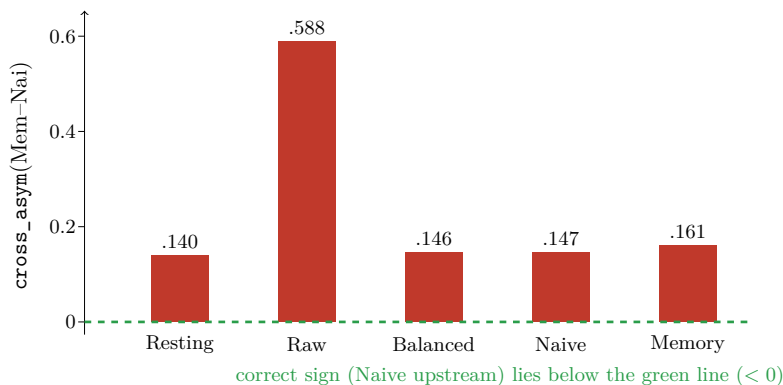


Figure 2: The inversion is control-independent. “Raw” is the ZERO control (no normalization = pure cross-engagement) and is the *most* positive, i.e. most inverted; every control keeps Memory above Naive. A genuine recovery would put these bars *below* the green line.

## 4 Takeaways

- **Boundary mapped.** `cross_asym` direction transfers to *temporal* / *disease-progression* axes (vaccination timeline here; COVID severity in §30) but is fundamentally mis-signed on a *cell-state differentiation* axis. The timeline is the right axis for an acquisition-based statistic; recovering *state* direction would need a retention/loss-keyed statistic.
- **Self-correction.** The control-swap decomposition falsified our own baseline-coincidence explanation — a reminder that the symmetric-control contrast and a raw (no-control) ablation are the right tools to attribute a direction result.
- **Engineering.** Staging the pipeline (inspect before GPU) caught three real schema issues up front: the atlas samples *Day 11* (not Day 10); obs fields are dotless (`celltype11/2`); and CCR7 is *absent* from the panel, so state gating uses CD45RO/CD27. The Seurat `.rds` was read without `SeuratObject` via base-R `attr()` slot access (only `Matrix` required).
- **Caveats.** Validation, not discovery (orders are textbook); cross-sectional (direction, not forecast); *early* (day-28) memory only; 6 donors (the bootstrap, not the point estimate, is what limits the timepoint verdict to AMBER); direction  $\neq$  causation.

Sources: `reports/vaccine_progression/` (VACCINE\_PROGRESSION\_RESULTS, STATE\_CONTROL\_DECOMP, PRE\_REGISTRATION); CLAUDE.md §32; cluster jobs 30907134–38.